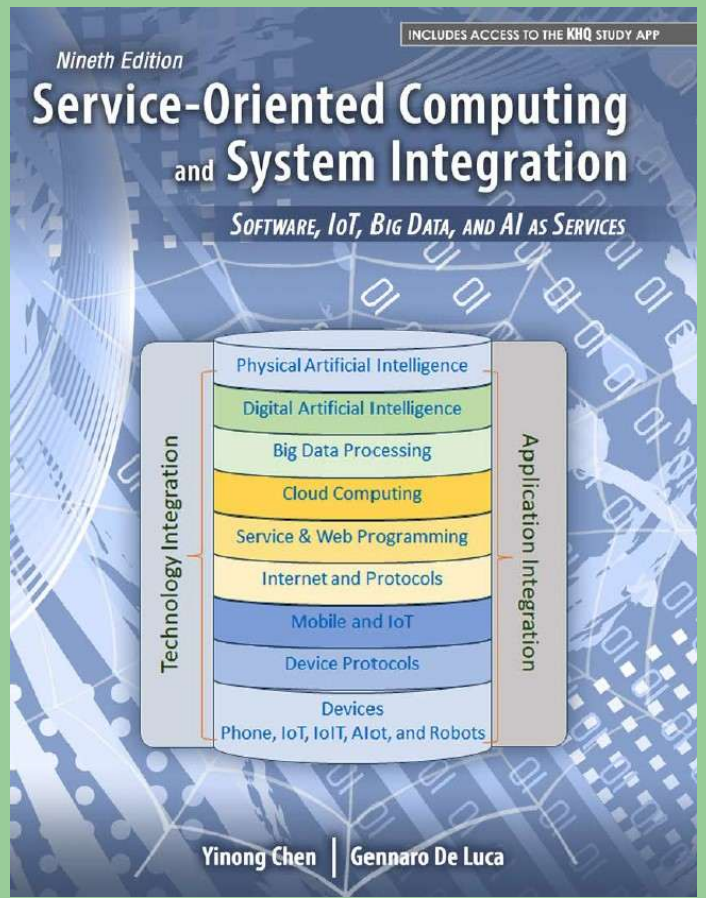
CSE446/598 Spring 2026

Software Integration and Engineering

Day One Itinerary: About the Course



Dr. Yinong Chen

https://myasucourses.asu.edu/

School of Computing and Augmented Intelligence

ASU IRA A. FULTON SCHOOLS OF engineering

Day One Itinerary

- About the instructor
- Course objectives & outcomes
- Syllabus discussion
- Course Introduction

Instructor: Yinong Chen

Joined ASU CSE in 2001

- This Spring:

  CSE445/598 Distributed Software Development
  CSE446/598 Software Integration and Engineering
  CSE551 Foundation of Algorithms
  FSE598 Emerging Computing Technologies

- Before this semester at ASU

  ASU 101, F11, F19
  CSE 101/FSE100: Every semester from Fall 06 to Fall 11, S16
  CSE 230: F11, F12, S13, F13, F14
  CSE 220: S2017, S2017, S2018, S2019
  CSE 240: Every semester since F2001
  CSE 310: SS01, F01, SS02
  CSE 225/EEE225: F02, S03, F03, S04; **CSE 330**: S02, SS 03 → **CSE230**
  CSE 420/598: S01
  CSE 423: (Capstone) S08, F09; CSE485 F15
  CSE 445/598: Almost every Fall and summer since 2006
  CSE 446/598: Every Spring and Summer since 2010.
  CSE 551 Foundations of Algorithms, Spring 2021, Spring 2023, Spring 2025, Spring 2026
  CSE 565 Software Verification, Validation, and Testing, F20, F22
  CSE 598 Foundations of Social Media Mining, Spring 2022
  FSE598 Emerging Computing Technologies, Fall 22, SP 23, SS 23, SP 24, F 24, SP 25

Yinong Chen

Before joining ASU

- Taught for six years

  School of Computer Science

  Wits University of Johannesburg, South Africa
- Postdoc at LAAS-CNRS, Toulouse, France
- Ph.D. from Karlsruhe Institute of Technology (KIT), Germany
- Contact and more …

  http://www.public.asu.edu/~ychen10/

Heinrich **Hertz** worked at KIT *from 1885 to 1888, where he discovered* electromagnetic waves

Yinong Chen

- More than 10 books
- 300 research papers, most are in service-oriented computing and computer science education
- Editor in Chief, Associate Editor, Editorial Board Member of international journals
- Chair of international conferences
- Keynote, panel talks
- Taught high school students to program robots in two semesters and also in summers

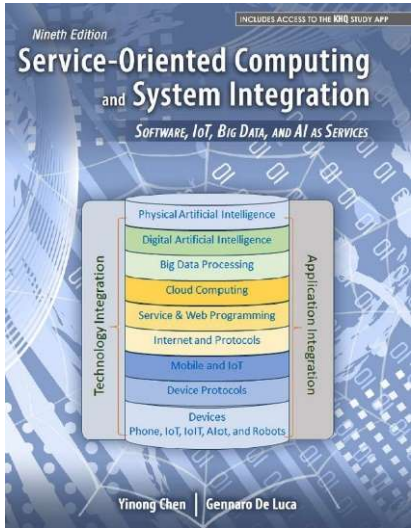
CSE446 SIE vs. CSE 445 DSD

CSE446 Software Integration and Engineering

Catalog Description

Software development using architecture design, composition, workflow, services, data resources, data representations, data management, and development tools.

Co-requisite: CSE445 (Does not apply to CSE598)



Key differences with CSE445

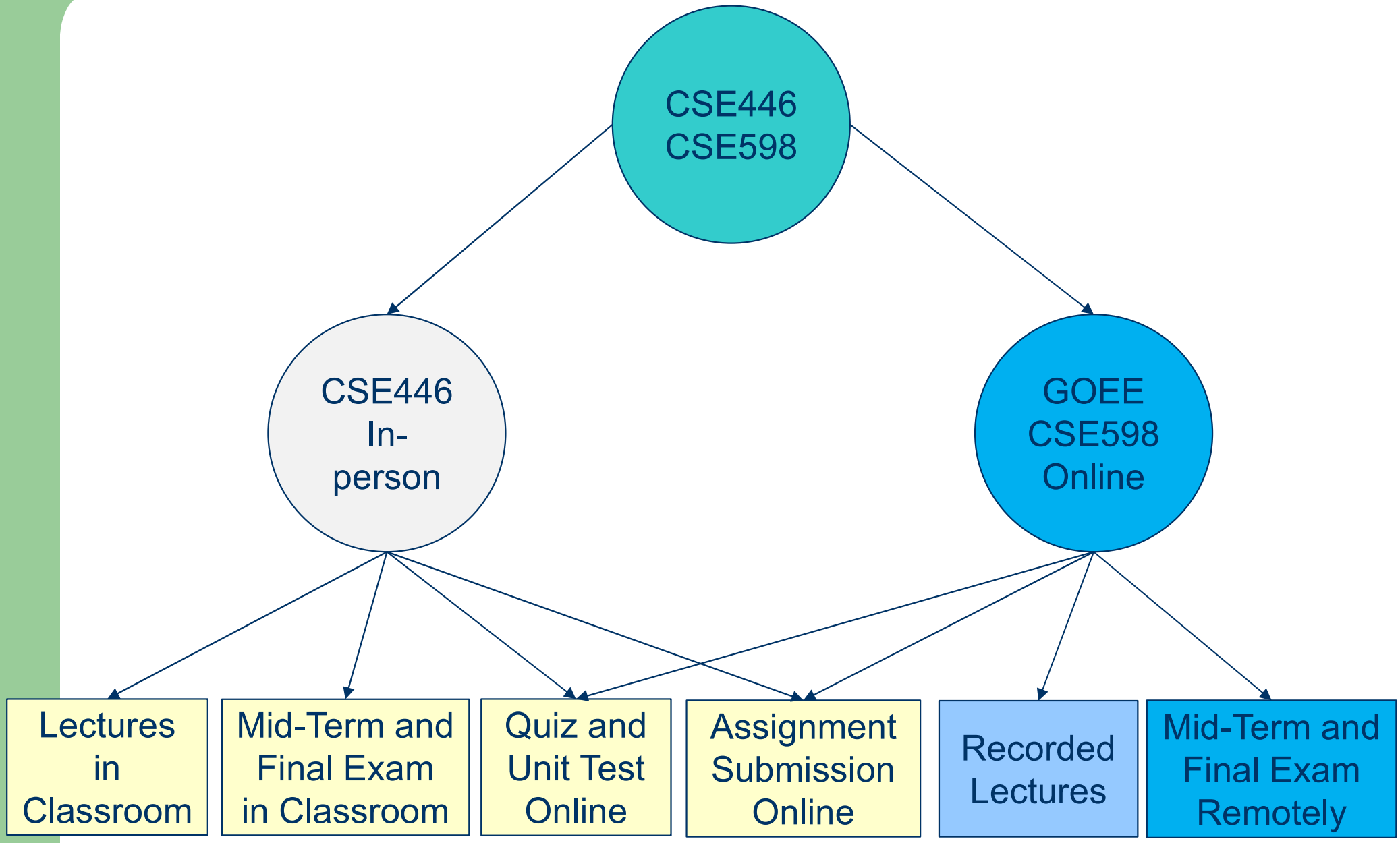
- Composition and workflow: architecture-driven software integration from existing components, AI, big data, cloud computing;
- Data sources and data management: Integration of computing with large data sources – databases and other data files, IoT, mobile, big data, and ontologies.

ASU Inclusive Text Access

- The Text will be provisioned as an e-book and made available at a discounted price, cheaper than if purchased directly from the publisher.
- If you wish to take advantage of this discounted group price, no additional action is needed. Following the drop/add period, the charge will post to your student account under the header "Digital Integrated Course Mtrl" and your access will continue uninterrupted.
- If you'd rather purchase the textbook from an alternate source, you may choose to opt out of the program by following the instruction in the syllabus.

Syllabus and Course Information

Sections: CSE446 and CSE598



Objectives and Outcomes

1. **To understand software architecture and software process**
   - Students understand the requirement and specification process in problem solving.
   - Students understand software life cycle and process management
   - Students can identify advantages and disadvantages of software architectures and their trade-offs in different applications

Standard objective and outcomes in any software engineering course

Objectives and Outcomes

2. **To understand and apply composition approach in software development**

- Students can apply software architecture to guide software development in the problem-solving process.
- Students understand interface requirement of software services.
- Students can compose software based on interfaces of services and components.
- Students can develop software system using different composition methods and tools.

Objectives and Outcomes

3. **To understand and apply data and information integration in software development**
   - Students can compose software systems using different data resources in different data formats.
   - Students can integrate application logic with different databases.
   - Students can apply the entire software life cycle to develop working software systems.

Weight and Grading Scale

Quizzes and unit tests will be given online for all students. You can take them in the lecture week.

Graded Activity	Weight
Assignments*	30%
Weekly Quizzes**	11%
Unit Tests 1, 2, 4, 5, 6*	12%
Mid-Term Exam	22%
Final Exam	25%

Percentage	Symbol Grade
96.5-100%	A+
92.5-96.4%	A
89.5-92.4%	A-
86.5-89.4%	B+
82.5-86.4%	B
79.5-82.4%	B-
75.5-79.4%	C+
69.5-75.4%	C
59.5-69.4%	D
Below 60%	E

No Y-Grade

* The lowest one will be dropped, no make up and no reset

** The lowest two will be dropped, no make up and no reset.

Canvas: Syllabus and Course Summary

- Home
- Announcements
- **Syllabus**
- Modules
- Assignments
- Discussions
- Grades

Course Syllabus

CSE446-598 SIE Syllabus.pdf and FAQforCourses445-446.pdf

Course-Calendar-446-598 Spring 2026.pdf

GradeInquiryForm.docx

Instructor: Yinong Chen

Instructor Yinong Chen Office Hours:

Office hours (Fall and Spring):
Monday, 9:15-11:30 (in-person and Zoom) Thursday: 9:15 to 11:30am (Zoom)
In-Person: BYENG M1-06 and Zoom: https://asu.zoom.us/j/7170231917

Teaching Assistant Office Hours:

TBD

Calendar File (PDF)

January						
Monday	Tuesday	Wedn	Thursday	Friday	Saturday	Sunday
12	13	14	15	16	17	18
Day 1 Issues Unit 1-1 Introduction		Unit 1-2 Self-Hosting Services				Quiz 0 for practice
19	20	21	22	23	24	25
Martin Luther King		Unit 1-3 Advanced WCF Services				
26	27	28	29	30	31	1-Feb
Unit 1-4 Unit 1 RESTful service 1		Unit 1-5 RESTful service 2			Assignment 1 Services Due	Unit 1 Quizzes & Unit Test Due

Quiz 0 Practice quiz

Assignment 1 Due

Unit 1 Quizzes and Test due on Mondays

Calendar File (PDF)

February						
Monday	Tuesday	Wedn	Thursday	Friday	Saturday	Sunday
						1 Unit 1 Quizzes & Unit Test Due
2 Unit 2-1 Advanced Web Application Architecture	3	4 Unit 2-2 Enterprise Software	5	6	7	8
9 Unit 2-3 Workflow	10	11 Unit 2-4 Power Platform	12	13	14 Assignment 2 Clients Due	15 Unit 2 Quizzes & Unit Test Due
16 Unit 3-1 Business Process	17	18 Unit 3-2 Business Process	19	20	21	22
23 Unit 3-3 Messaging	24	25 Unit 3-4 Web Caching	26	27	28 Assignment 3 Workflow and Messaging Due	1 Unit 3 Qui 1 $ Quiz 2 Due

Assignments Due

quizzes and Unit Test due on Mondays

Calendar File (PDF)

March						
Monday	Tuesday	Wedn	Thursday	Friday	Saturday	Sunday
						1 Unit 3 Qui 1 & Quiz 2 Due
2 Unit 4-1 VIPLE Workflow	3	4 Mid-Term Exam in Coor 170	5	6	7	8
9 Spring	10 Break	11 Spring	12 Break	13 Spring	14 Break	15 Sprng
16 Unit 4-2 VIPLE Advanced	17	18 Unit 4-3 Mobile Computing	19	20	21	22
23 Unit 4-4 IoT and Device Integration	24	25 Unit 5-1 ADO	26	27	28 Assignment 4 VIPLE and Phone Due	29 Unit 4 Quizzes & Unit Test Due
30 Unit 5-2 LINQ to Object	31					

Mid-Term Exam

quizzes and Unit Test due on Mondays

Assignments Due

Calendar File (PDF)

April

Monday	Tuesday	Wedn	Thursday	Friday	Saturday	Sunday
		1 Unit 5-3 LINQ to SQL	2	3	4 Unit 5 Quiz 1 available	5 Course Withdrawal Deadline
6 Unit 5-4 LINQ to XML	7	8 Unit 6-1 Big Data Concept	9	10	11 Assignment 5 LINQ Due	12 Unit 5 Quizzes & Unit Test Due
13 Unit 6-2 Big Data Hadoop	14	15 Unit 6-3 Big Data Python	16	17	18	19
20 Unit 6-4 Artificial Intelligence	21	22 Unit 6-5 Ontology	23	24	25 Assignment 6 Big Data Due	26 Unit 6 Quizzes & Unit Test Due
27 Unit 7-1 Cloud Computing	28	29 Unit 7-2 Cl Computi				

May

Mon	Tuesday	Wedn	Thursday	Friday	Saturday	Sunday
				1	2	3 Unit 7 Quiz 1 Due
4 CSE446 Final Exam scheduled by University 12:10 - 2:00 PM in PSF 166	5	6	7	8	9	10

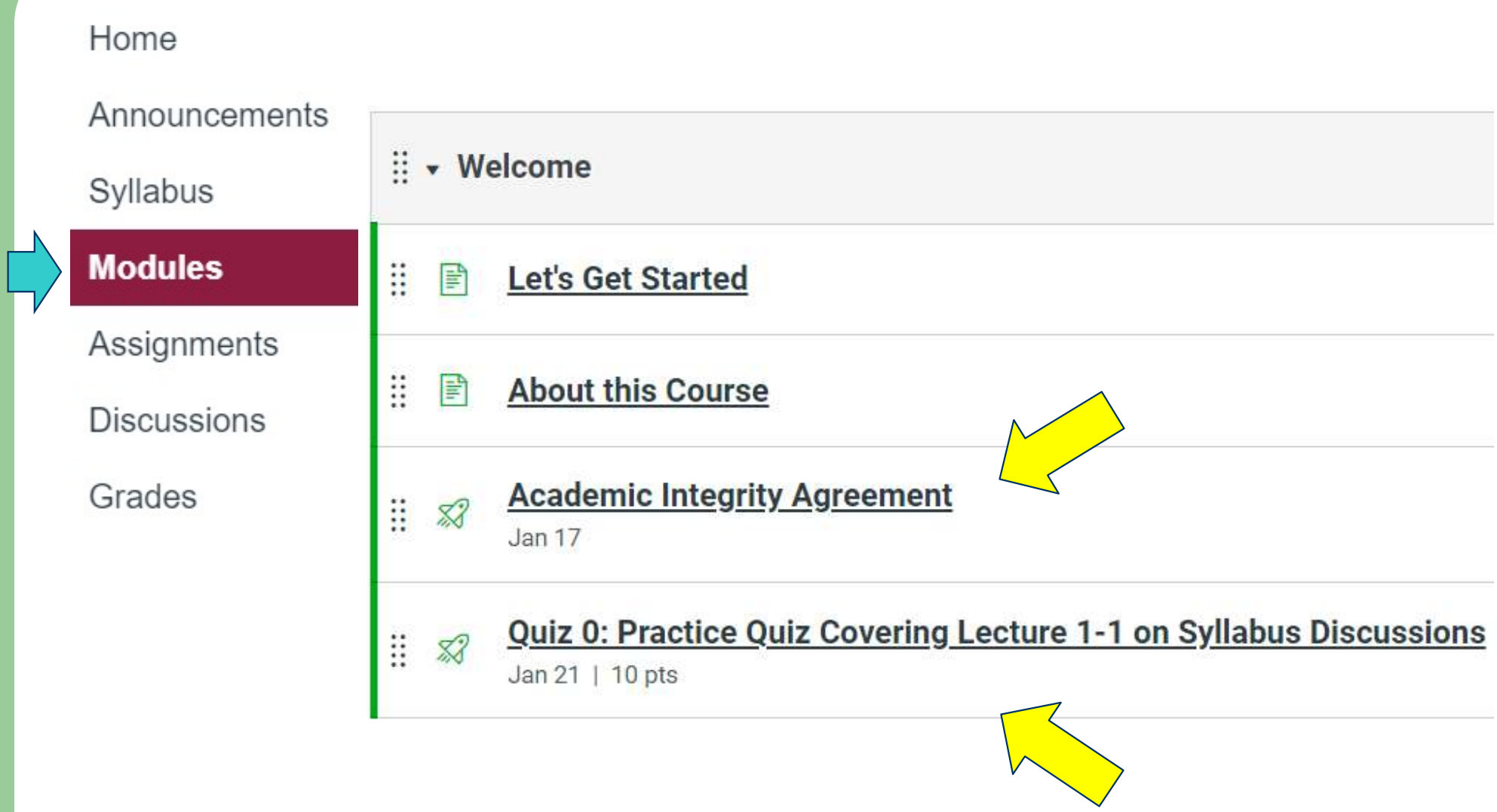
Final Exam Schedule: https://students.asu.edu/final-exam-schedule

Canvas: Modules and Online Lectures

- Home
- Announcements
- Syllabus
- **Modules**
- Assignments
- Discussions
- Grades

- Welcome
- Unit 1: Introduction and Advanced Service Development
- Unit 2: Software Architecture and Integration
- Unit 3: Business Process Execution and Integration
- Unit 4: Device Integration and Mobile Computing
- Unit 5: Web Application and Data Integration

Canvas: Modules and Online Lectures



Canvas: Modules = Units

Home

Announcements

Syllabus

Modules

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Grades

Unit 1: Introduction and Advanced Service Development

INTRODUCTION

Unit 1 Introduction

LEARNING MATERIALS

Unit 1: Read / Lecture Slides and Videos

TESTS and ASSESSMENTS

Unit 1 Discussions / Community Forum on Service Standards and Service D

Unit 1 Quiz 1: covers Lectures 1-1, 1-2 and 1-3
May 27 | 10 pts

Unit 1 Quiz 2: covers Lectures 1-4 and 1-5 on RESTful Services
May 27 | 10 pts

Unit 1 Test: Lectures Unit 1-1, 1-2, 1-3, 1-4, and 1-5
May 27 | 20 pts

ASSIGNMENT: Note the Due Date!

A1 Adv Service Submission

There are 7 units

Each unit has 4 or 5 lectures

Each unit has 2 quizzes

Each unit has 1 unit test

Each unit has 1 assignment

Canvas: Discussion Boards

Home

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ASU Course Policies

Unit 1 Discussions / Community Forum
All Sections

Unit 2 Discussions / Community Forum
All Sections

Unit 3 Discussions / Community Forum
All Sections

Unit 4 Discussions / Community Forum
All Sections

Each unit has a discussion forum

Published | Edit

Unit 1 Discussions / Community Forum
Yinong Chen
All Sections
Nov 3, 2021 at 4:48pm

Search entries or author | Unread | Subscribed

Reply

Read FAQ document in Course Web

Frequently Asked Questions and Answers

Attached Files: FQAforCourses445-446.pdf (210.154 KB)

Q: If I missed the deadline for taking an online test, can I make up the missed test?
A: No. Online tests may not be taken after the due date. However, the two lowest scores of the quizzes AND the lowest score of the chapter/unit tests will be dropped. If you missed two quizzes and one chapter/unit test, the zero grades will be dropped automatically in calculating the weighted average grade.

Q: If I missed the deadline for taking online quizzes/tests, can I get their solutions for study purposes?
A: No. Online test questions are generated automatically from a pool of questions. If you do not take them, they are not generated. Therefore, do not miss online quizzes/tests! You can use the sample questions and solutions in the textbook for study purposes.

Q: I need extra credit to improve my grade. Can you please give me extra assignments for the extra credit?

A: No extra credit-activities will be given to any individual. Extra credit-activities are given to the entire class through dropping one chapter/unit test and two quizzes.

As written in the syllabus, an alternative to the assignment and exam may be arranged if a student misses the activity and the absence is caused by documented illness or personal emergency that made the completion/attending impossible. A written explanation (including supporting documentation) must be submitted to the instructor before the part of work is due or as soon as the circumstances are known.

Q: I missed 1% point to receive a B (for example) grade. This is important to me and to my family. If I do not receive a B in this course, I will lose my scholarship, and I have to drop from the university. Can you please take this situation into account to move my grade up?

Standard Classroom Expectation

- Silent your cellular phone; If your phone happens to ring, stop it immediately and do not answer your phone!
- Use computer for directly related activities only, e.g., taking notes during the lectures.
- Do not talk to each other during the lecture. If you have a question that needs to be resolved immediately, you must ask the instructor.
- Enter the classroom before the lecture's starting time.
- Do not leave the classroom during the lecture, unless there is an emergency situation.

Policies

- Interaction: You are encouraged to ask the instructor questions during the lectures.
- Outside class help welcome and encouraged:
  - Discussion board (effective and fair);
  - Instructor's and the TA's office hours;
  - Request appointments if you can not make the office hours;
  - Email/Canvas mail for personal issues
- Tests and exams: Missing tests and exams will be giving zero credit and may not be made up.
- Assignments: Late submission will be accepted with grade deduction: Please check assignment document for details.

Extra credit, alternative, and inquires

- No extra credit-activities will be given to any individual. Extra credit-activities may be given to the entire class.
- An alternative to a graded activity may be arranged if a student's absence is caused by **documented illness or personal emergency**. A written explanation (including supporting documentation) must be submitted to the instructor before the part of work is due or as soon as the circumstances are known.
- Any inquires or appeals on grades of homework, projects, or tests must be done in writing by completing the "Grade Inquiry Form" within a week from the day the grades and/or comments were published on-line. State the problem and the rationale for any change in grade in your appeal.

Cooperation and Code of Conduct

- You are encouraged to cooperate in study group on learning course materials.
- You may not cooperate on preparing the individual assignments**.** Anything you turn in must be your own work. If you use an idea that is found in a book or other sources, make sure you acknowledge the source and/or the names of the persons in the write-up for each problem.
- All assignment questions must be asked in the course discussion board. Asking assignment questions or making your assignment available in the public websites before the assignment due will be considered AIP (Academic Integrity Policy) violation.
- The instructor and the TA/grader are required to CAREFULLY check any possible proliferation or plagiarism. We will use the software tools like MOSS (Measure Of Software Similarity) to check any assignment. The university expects all students to adhere to ASU's policy on Academic Dishonesty. These policies can be found in the Code of Student Conduct:

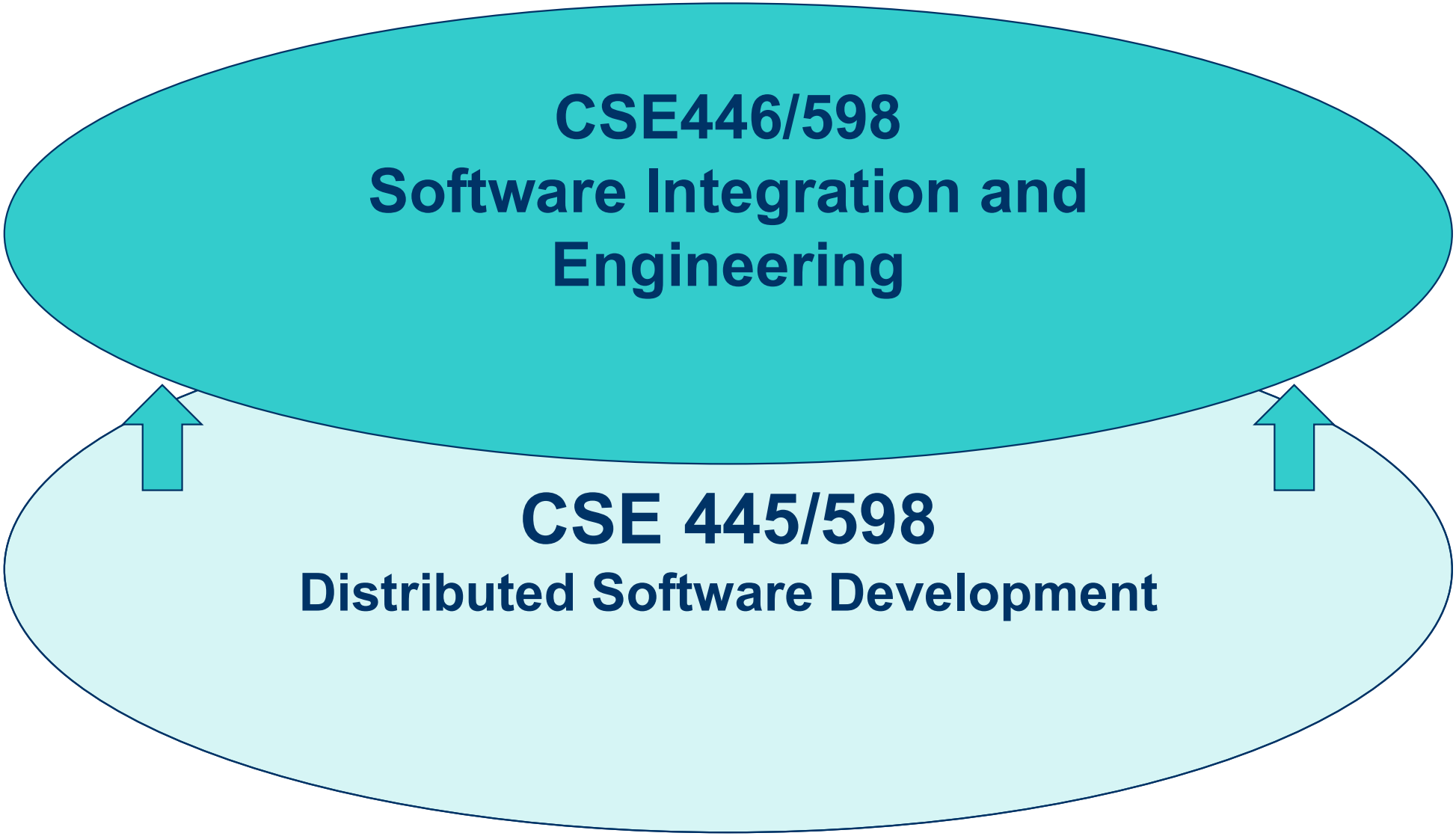
  https://provost.asu.edu/academicintegrity/policy

  ALL cases of AIP violation will be handed to the AIP office. Penalties include a failing grade in the class, a note on your official transcript (XE) that shows you were punished for AIP violation .

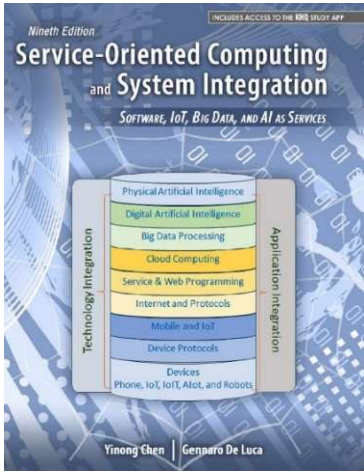
Announcement and Information

- Official announcements will be made either in the course "Announcement" part of the course web page.
- Articles or answers in the discussion board by the instructor, the TA, or other students are not official announcement.
- Make sure you regularly (at least once every two days) check the course web page for any announcements.

CSE 445/598 (DSD) vs. CSE446/598 (SIE)



One Text for Two Courses



DSD

Part I	Distributed Service-Oriented Software Development and Web Data Management
Chapter 1	Introduction to Distributed Service-Oriented Computing
Chapter 2	Distributed Computing with Multithreading
Chapter 3	Essentials in Service-Oriented Software Development
Chapter 4	XML and Web Data Formats
Chapter 5	Web Application and Data Management
Chapter 6	Dependability of Service-Oriented Software

Reading

Assignment 1

SIE

Part II	Advanced Service-Oriented Computing and Large System Composition
Chapter 7	Service-Oriented and REST Architecture
Chapter 8	Enterprise Software Development and Integration
Chapter 9	IoT, Robotics, and Device Integration via Visual Programming
Chapter 10	Interfacing Service-Oriented Software with Database
Chapter 11	Big Data and Cloud Computing
Chapter 12	Artificial Intelligence and Machine Learning
Chapter 13	Mobile Computing and Application Development

Topics/Units To be Covered

- Unit 1 - Chapter 7:
  Introduction, Advanced SOA and REST Architecture
- Unit 2 - Chapters 7 and 8:
  Software Integration: Architecture, Workflow, Copilot
- Unit 3 - Chapter 8:
  Software Integration: Antigravity, BPEL, Messaging & Caching
- Unit 4 - Chapters 9 and 13
  IoT, Robotics, and Mobile Computing
- Unit 5 - Chapter 10:
  Web Application and Data Integration
- Unit 6 - Chapters 11 and 12:
  Big Data and Ontology, AI, and Software as a Service
- Unit 7 – Chapter 11
  Cloud Computing

Unit 1

Advance SOA and REST Architecture

1. Day 1 Issues and Introduction
2. Self-Hosting Services
3. Advanced WSDL/SOAP Services, including asynchronous services
4. REST Concepts
5. RESTful Services and Web APIs

Unit 2

Software Integration: Architecture and Workflow

1. Advanced Web Application Architecture
2. Enterprise Application Architecture
3. Workflow Concepts and Workflow Foundation
4. Power Platform:
    1. Power Apps,
    2. Power Automate,
    3. Power BI,
    4. Power Pages, and
    5. Copilot Studio

Unit 3

Software Integration: LLM, BPEL, Messaging and Caching

1. Google Antigravity: AI-LLM-Based Development
2. BPEL Process
3. Message-Based Integration
4. Web Caching and Recommendation

Unit 4

IoT and Mobile App Development and Device Integration

1. Visual IoT/Robotics Programming Language Environment
2. Advanced VIPLE Workflow and Application
3. Device Integration
   - Internet of Things
   - Service-Oriented Robotics Computing
   - Event-Driven Robotics Applications
   - Robot as a Service in Cloud Computing
4. Mobile Computing and Phone App Development, with synchronous and asynchronous service accesses.
   - iOS phone
   - Android phone
   - Android phone simulator

Unit 5

Application and Data Integration

1. ADO for Database Access
2. LINQ to Object
3. LINQ to SQL and Functional Programming
4. LINQ to XML

Unit 6 Big Data and AI

1. Big Data Concepts and Domains
   - Concepts and Infrastructure
2. Big Data Processing on Hadoop
   - MapReduce, Hadoop, and Analytics
3. Big Data Processing in Python
4. Introduction to AI and Machine Learning
   - AI Development and Domains
   - Case Study: Image recognition
   - Case Study: Flight pattern training and recognition
5. Ontology for AI and Semantic Web
   - Ontology Languages RDF and RDF Schema
   - Web Ontology Languages and IDEs

Unit 7

1. Cloud Computing and Software as a Service
   - Software as a Service, Platform as a Service, and Infrastructure as a Service,
   - Multi-tenancy
2. Cloud Computing Case Studies
   - Google Cloud: from concepts to implementation
   - Microsoft Azure from development to deployment
   - Oracle Cloud
   - Amazon Cloud

School of Computing and Augmented Intelligence
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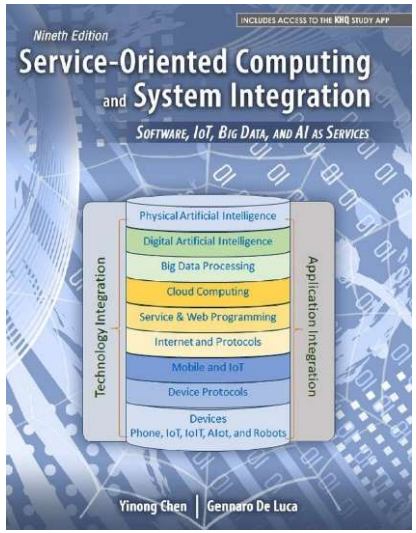


CSE 446 / CSE 598

Software Integration and Engineering

Unit 1

Service Standards and Service Development

Lecture 1-1

Introduction

Dr. Yinong Chen
https://myasucourses.asu.edu/

Roadmap of Unit 1

1. Introduction
   - **Software Integration**
   - **Service-Oriented Architecture**
   - **Service-Oriented Computing**
   - **Service-Oriented Development**
   - **Semantic Web and Intelligent Web**
2. Self-Hosting Services
3. Advanced WSDL/SOAP Services
4. REST Concepts
5. RESTful Services

What is a System?

A **system** is a collection of interconnected components that have some functions or purposes. A system can be defined at three levels

- A system has **architecture**
  - the overall organization of its components
- A system has **interfaces**
  - input/output
- A system has **behaviors**
  - declarative relationship between input and output;
  - procedural relationship: transformation from input to output.

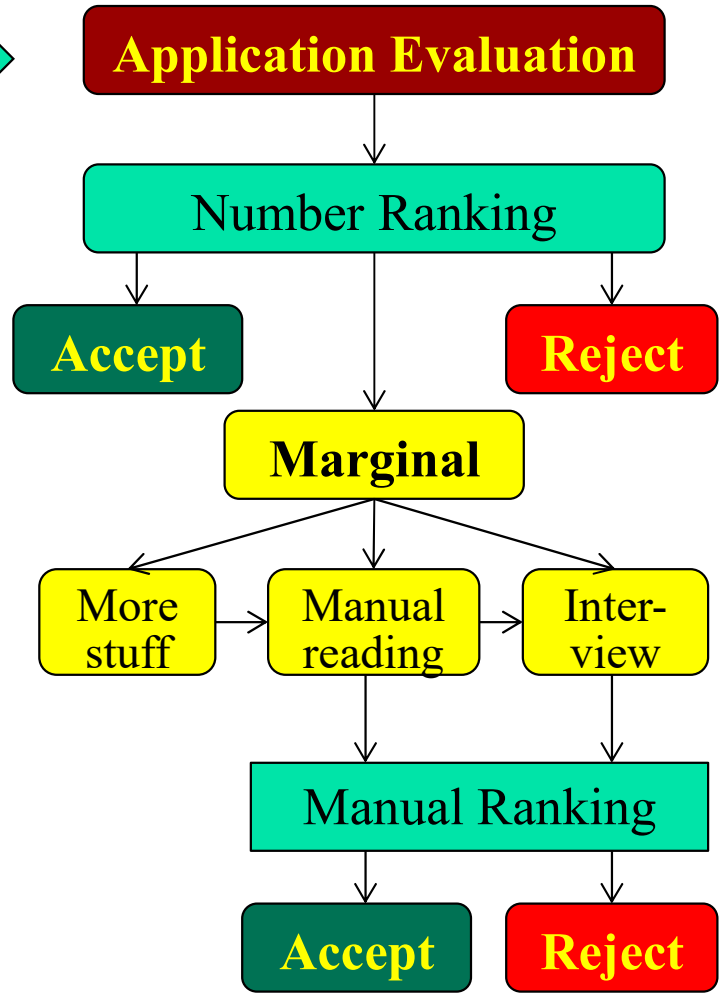
Software Integration at Different Levels

- **Architecture**
  - Integration of systems
  - Integration of components
  - Integration of computing systems and data sources

  The distinction may or may not be clear, depending on the context.
- **Interfaces**
  - Integration based on component models
    - *Object Model (migration or remote objects)*
    - *Service Model*
  - Integration based on communication model
    - *Point to point (Remote procedure call and invocation)*
    - *Bus (object broker and service bus)*
- **Behaviors**
  - Integration based on declarative relationship among inputs and outputs: Integration through message exchange among components (services and objects).
  - Integration based on workflow among the components

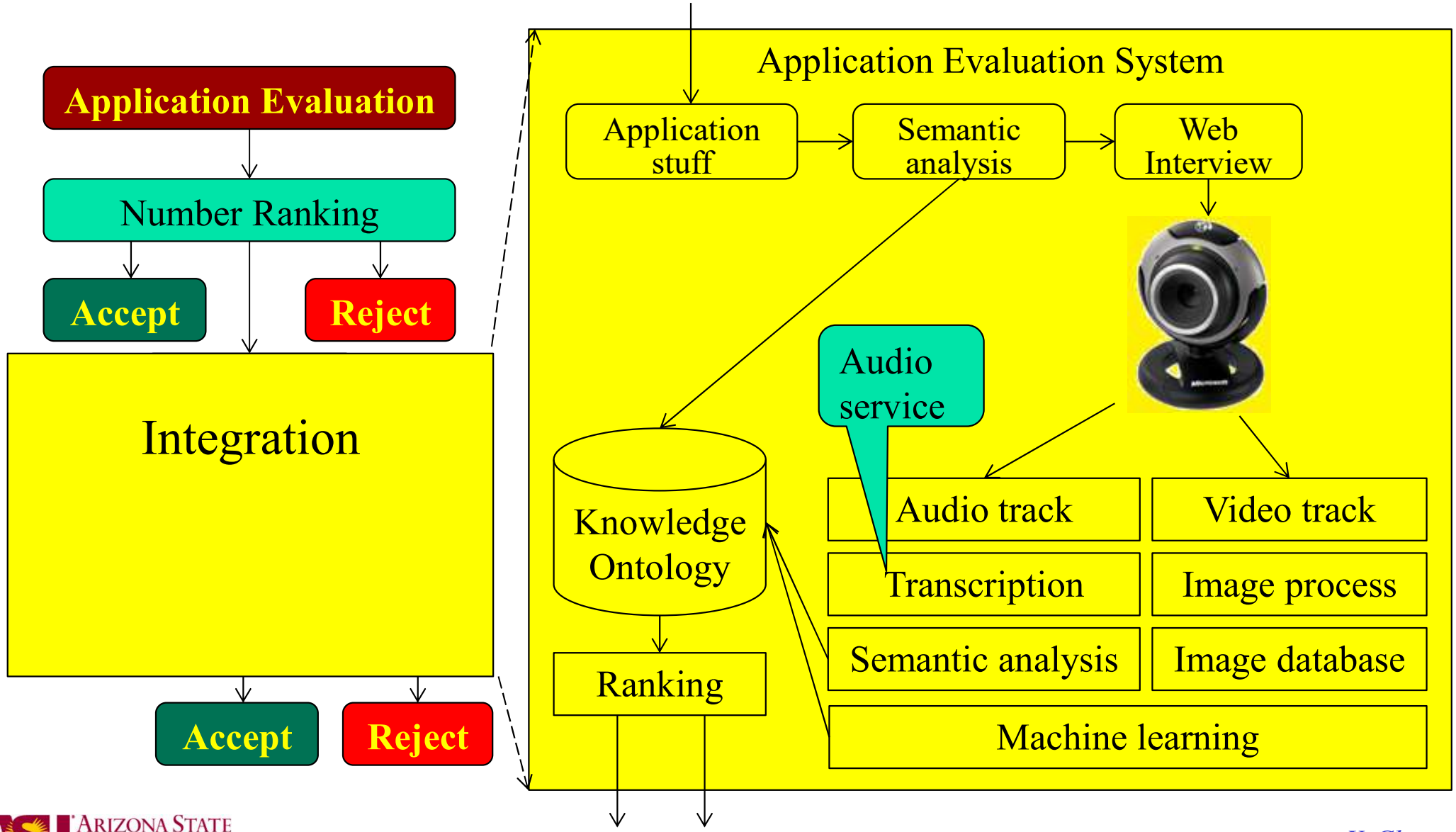
Integration Case Study: ASU Website

- Enrollment
  - Application submission
  - Application evaluation
- My ASU: Canvas
  - Class search and enrollment
  - Class schedule
  - Test and assignment submission
  - Grade book
  - Mail and alert
- Student database
- ASU Mail
  - Mail
  - Calendar and schedule



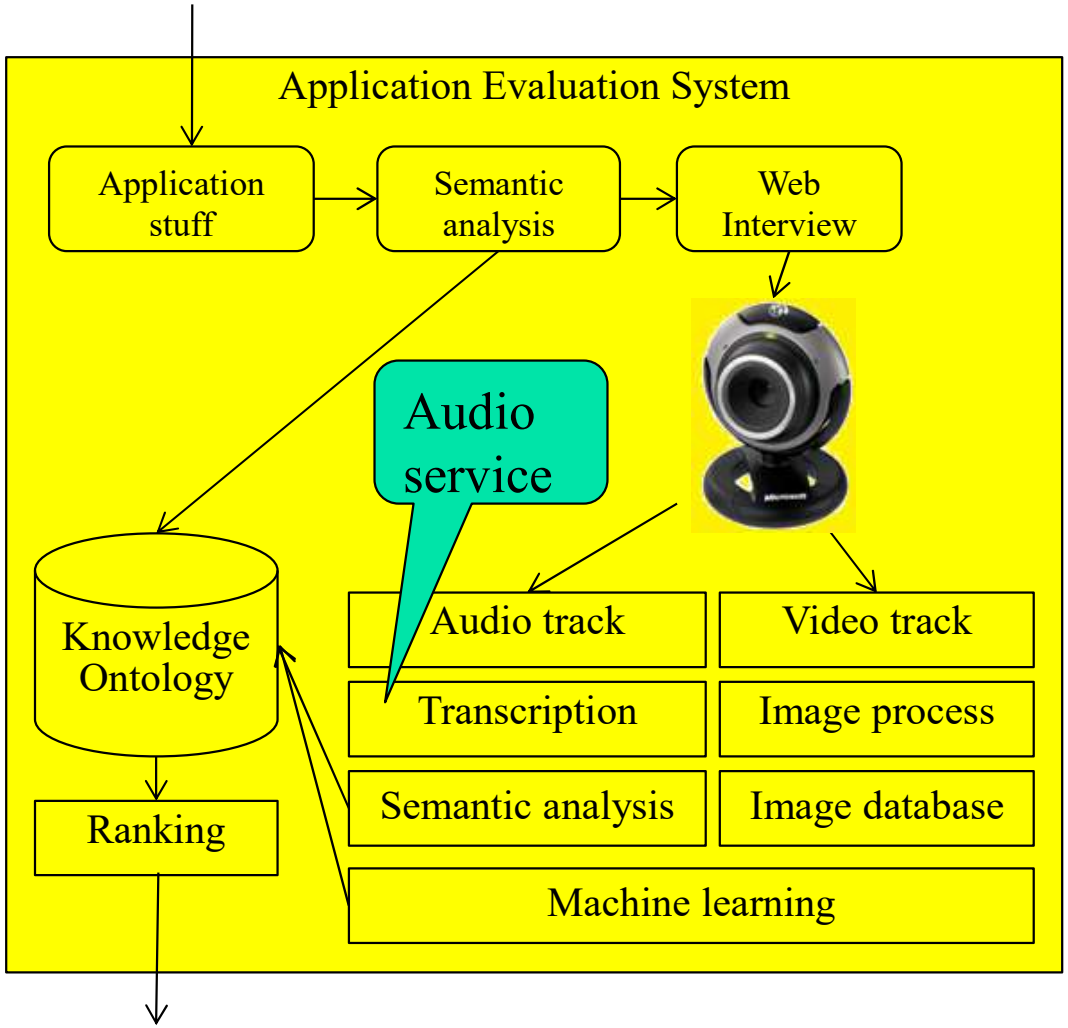
Integration Case Study: ASU Website

What else can we do to improve our business?



Starting Your Own Business

What can you do once you have developed the system?



- Start your own HR business to help any company to do prescreening;
- Integrate your system into any organization's IT system;
- Make each component available as services to integrate into other system, e.g., your ontology, transcription service, or image processing;
- Deploy your system into a cloud environment, so that you can focus on improving your business logic, instead of working on hosting business.
- What is the Amazon model?

Definitions and Terminologies

- **Service Orientation** (SO): Umbrella concept for all service-oriented approaches, including SOA, SOC, and SOD.
- **Service-Oriented Architecture** (SOA): Software consisting of a collection of loosely coupled and platform-independent services that interact with each other through standard interfaces. SOA does not concern developing operational software.
- **Service-Oriented Computing** (SOC) refers to the paradigm that represents computation in SOA. SOC concerns interface and behavior, including communication protocols, algorithms, and data representations.

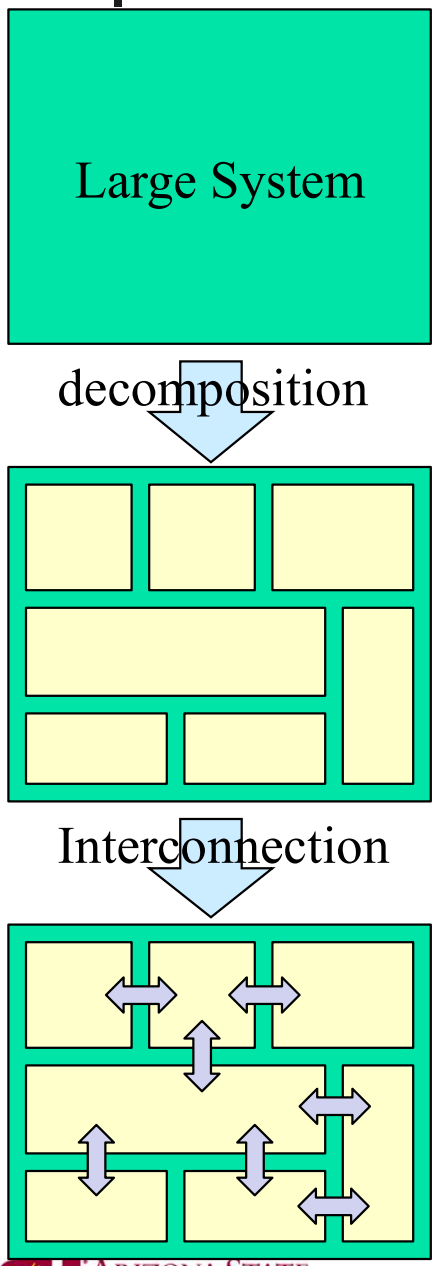
  SOA and SOC can be used alternatively if the entire system is concerned, without specifically referring to architecture, interface, or behavior.

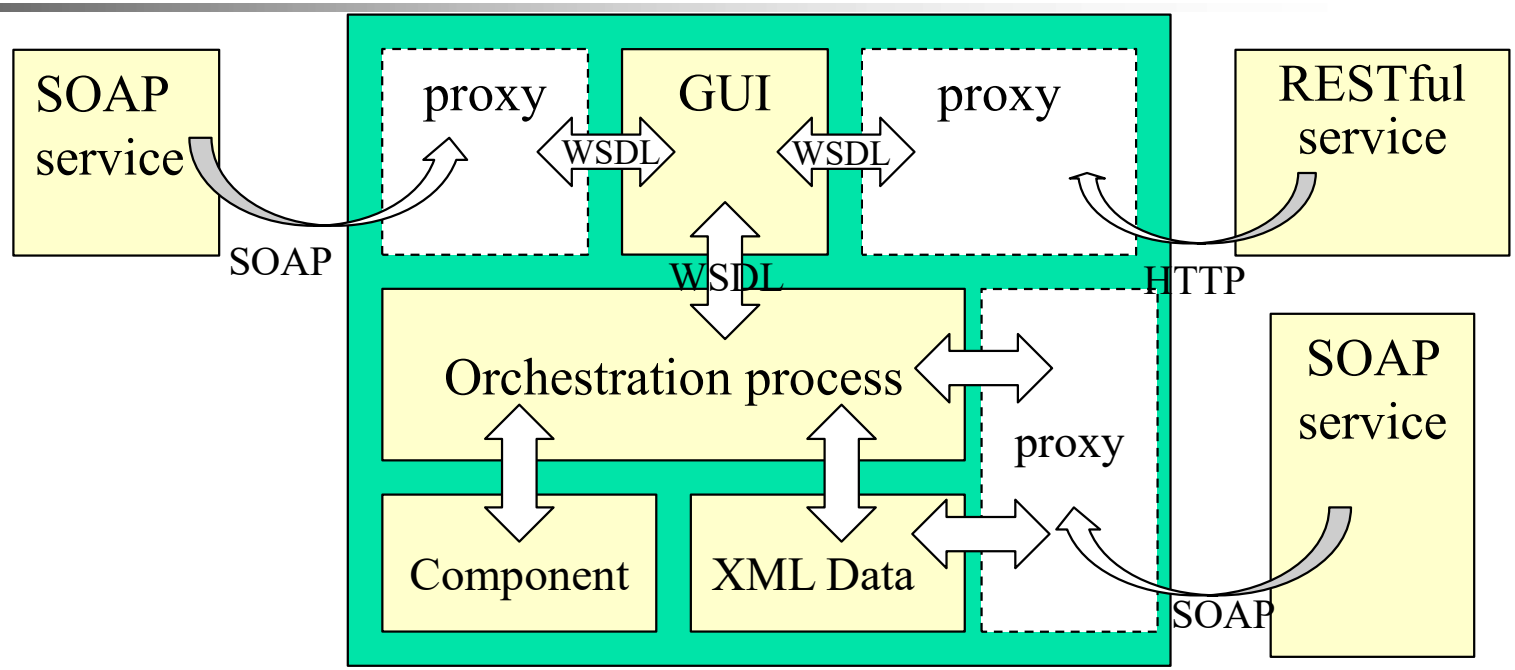
Definitions and Terminologies (contd.)

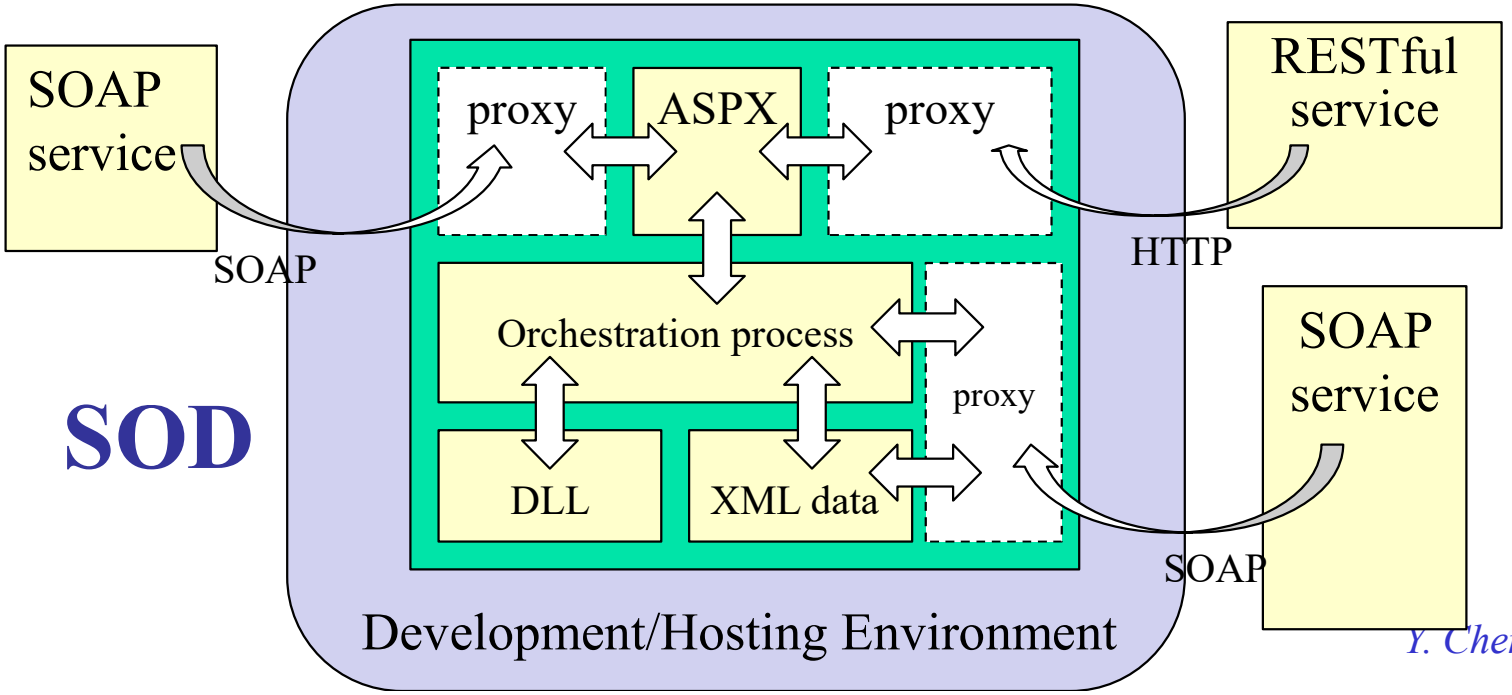
- **Service-Oriented Development** (SOD)
  - concerns the entire software **development life cycle** based on SOA concepts and SOC paradigm.
  - involves current technologies and tools to effectively produce operational software:
    - Programming language features,
    - Development environments: VS, Java EE, WebSphere;
    - Standards and Interfaces with components, services, and data sources;
    - Deployment and hosting environment: Self-Hosting, Servers, AWS hosting, Cloud Infrastructure, such as Google App Engine, MS Azure, etc.

SOA

SOC (adding behavior & Interface)







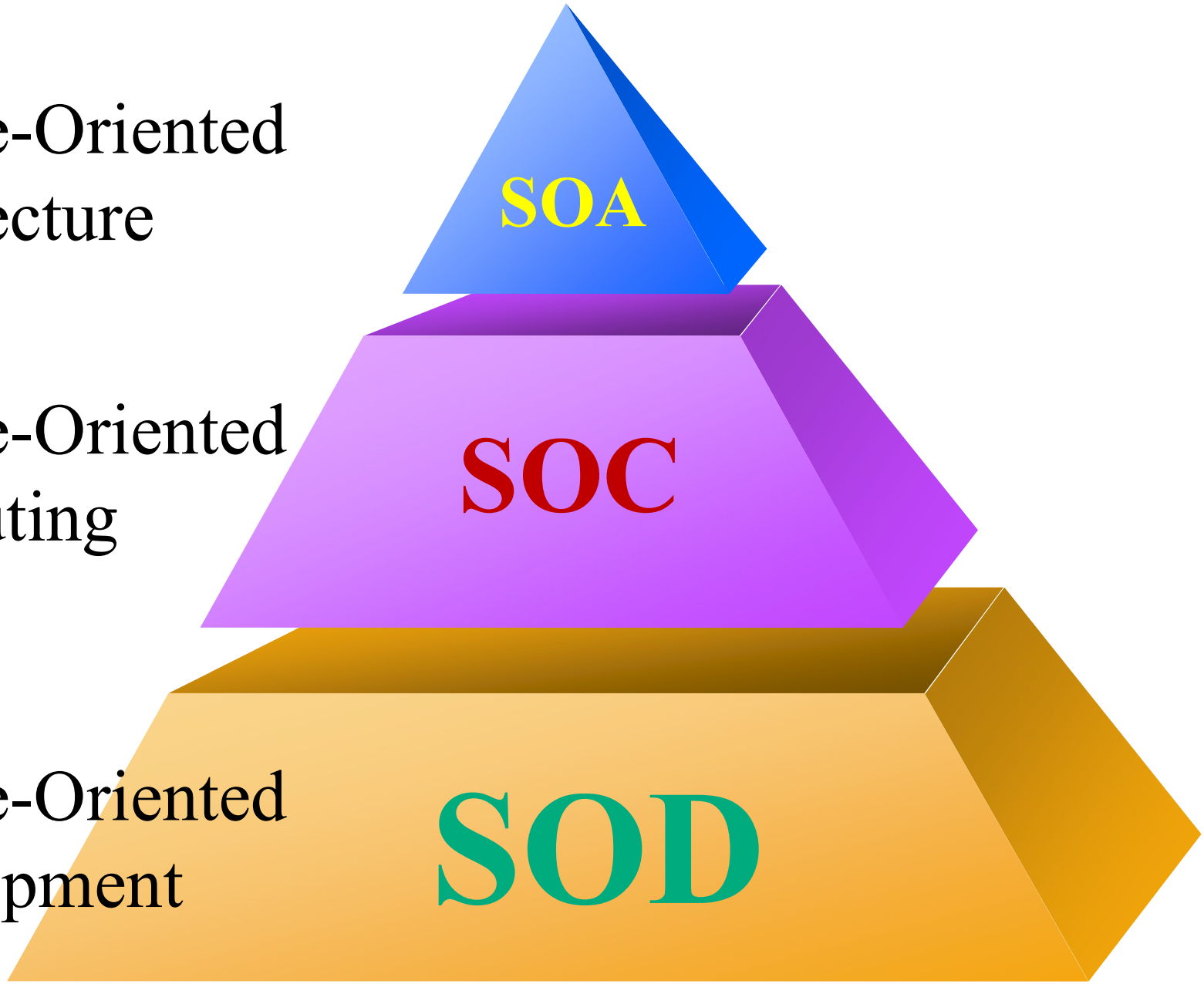
SOD

Service Orientation (SO)

Service-Oriented Architecture

Service-Oriented Computing

Service-Oriented Development



Why Service Orientation?

- Service orientation is similar to object orientation
- There is little difference at architecture level
- Driven by Development Requirements (developing operational software):
  - Large scalable and evolvable enterprise systems
  - Integration among existing systems
  - Interoperable among languages and platforms
  - Manageable complexity
  - Reusable business functions
  - Internet (Web) provisioning of services
  - Reduced cost of doing business
  - Increased dependability (reliability, availability, and security)

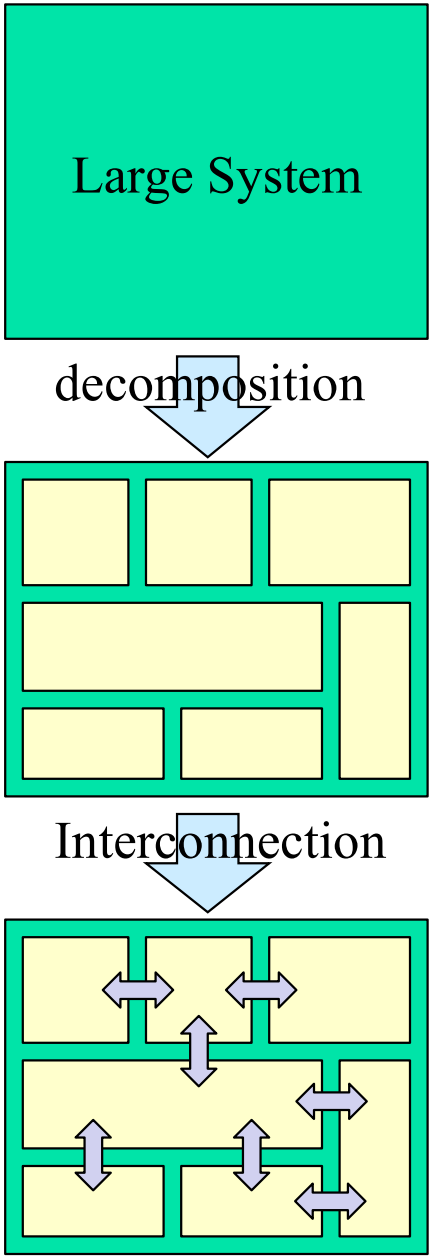
Why is SO Different from Object Orientation? Interface and Environment!

- OO has evolved into distributed object systems
  - CORBA, COM, DCOM, RMI (remote method invocation), etc. all try to implement *transparent* distributed systems
  - Standard interface such as IDL in CORBA
- SO standards are jointly developed by major players
  - IBM, Microsoft, Oracle, SAP, Sun Microsystems, …
  - SOAP, WSDL, HTTP, UDDI, URI
- Loosely coupling instead of tightly coupling
  - Message exchange instead of remote calls
  - Using proxy instead of code integration or migration
- Infrastructure for Service Publication (Service Brokers)
  - OO does not have standard publication mechanisms
  - SO enable and encourage sharing through standard brokers

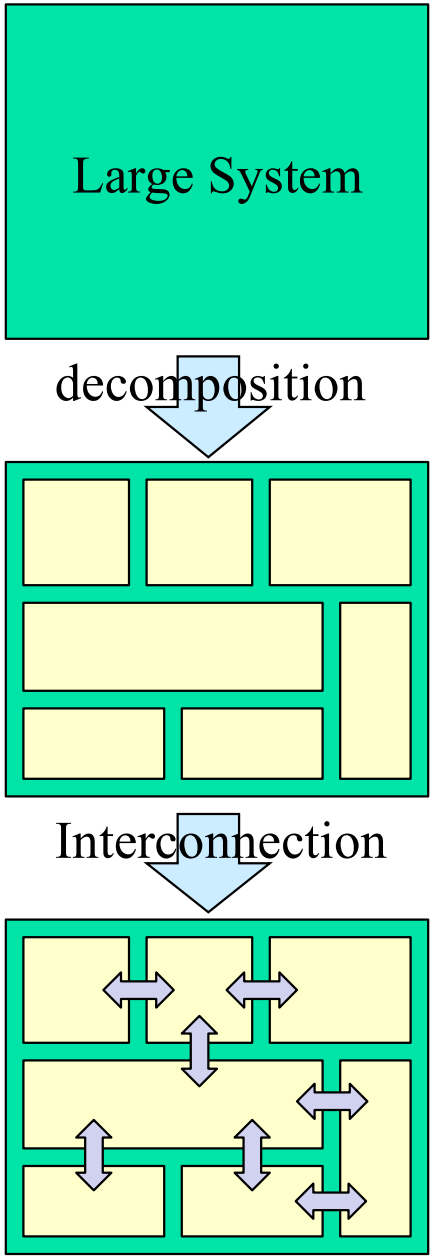
Better for integration

SOA vs. Object-Oriented Architecture (OOA)

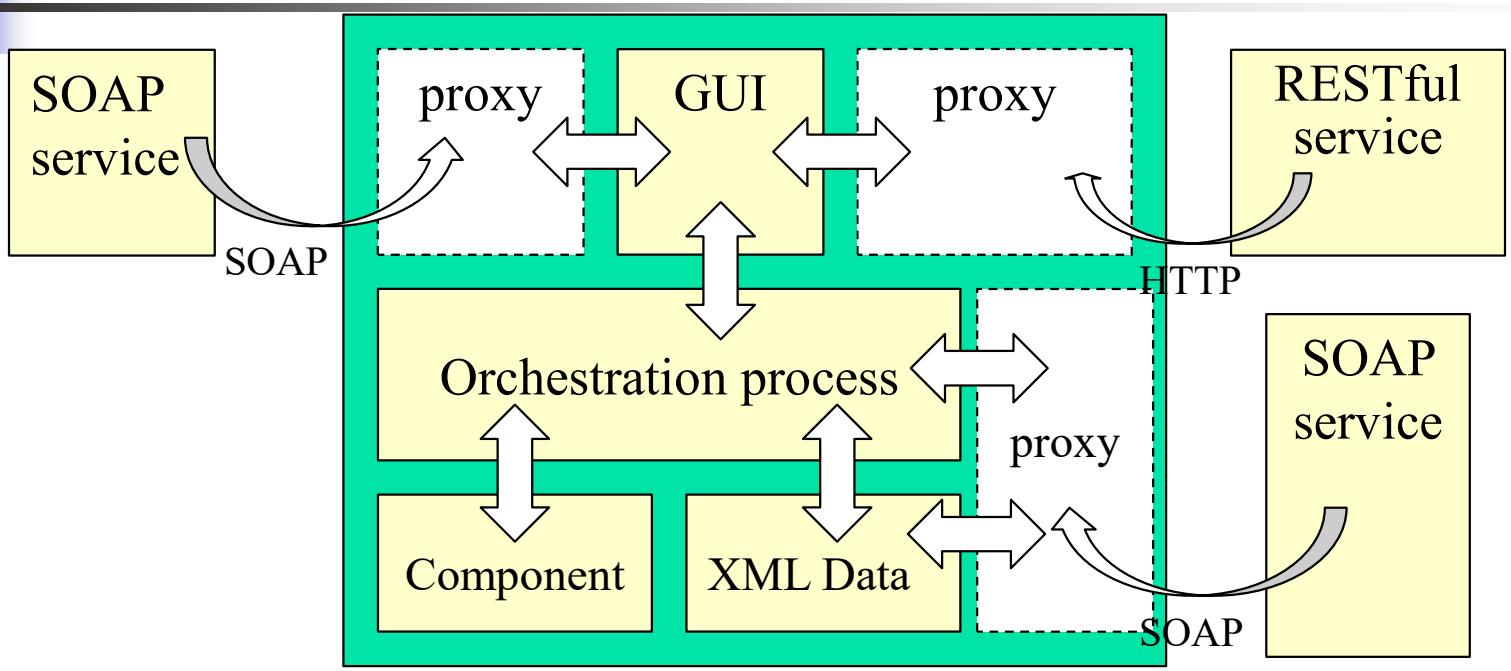
SOA

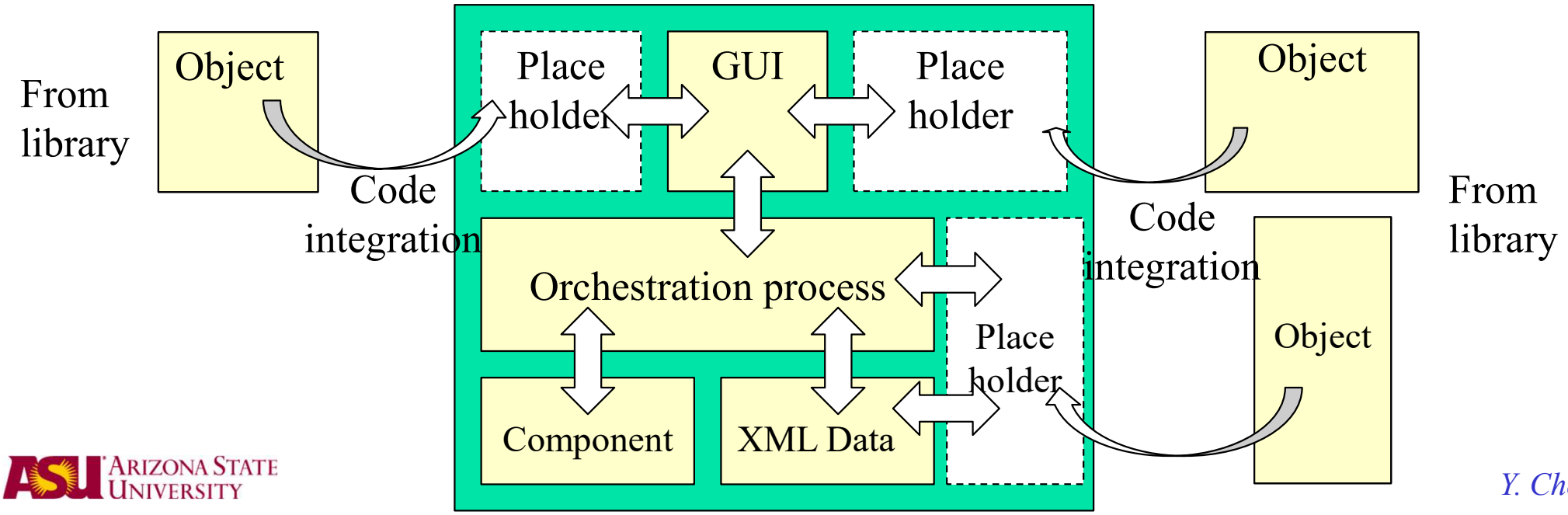


OOA

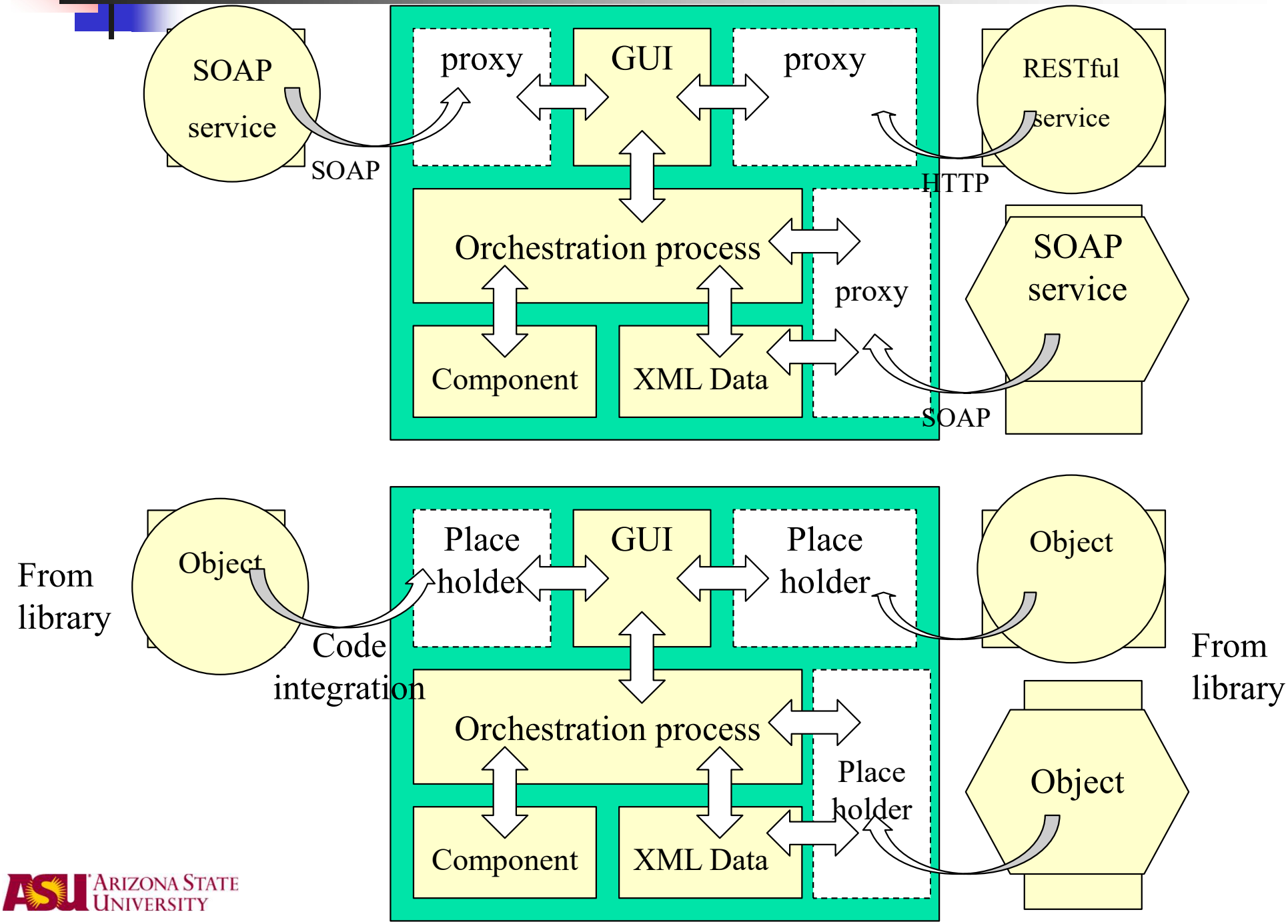


Interface & Behavior Levels: SOC vs. OOC





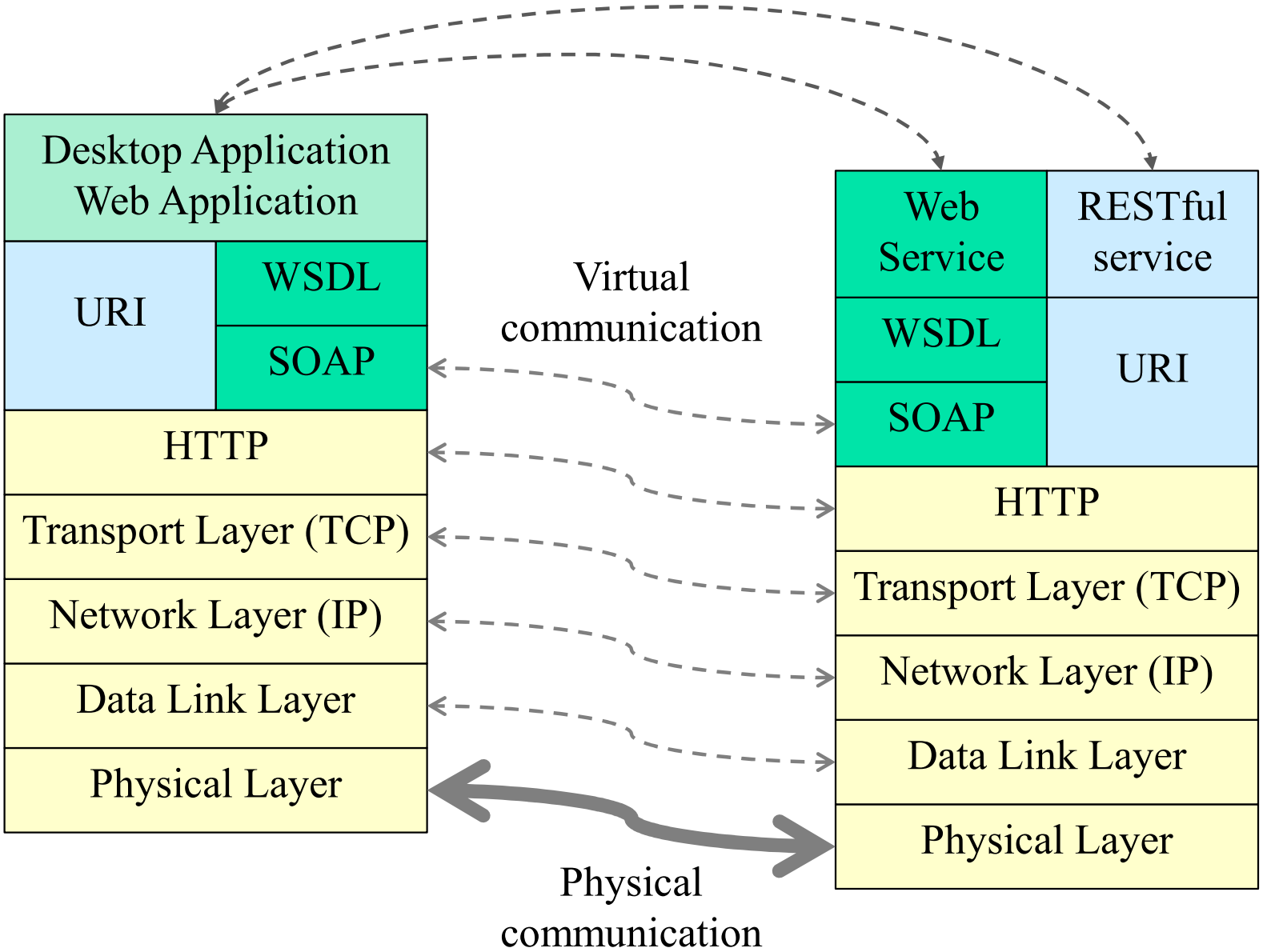
Interface & Behavior Levels: SOC vs. OOC



Definition of Web Service and SOAP Service

- A Web service is an **interface** that describes a **collection of operations** (Web methods) that are Web-accessible through standardized XML messaging
  - WSDL (Web Service Description Language) is used to describe the interface;
  - SOAP (Simple Object Access Protocol) is used to exchange messages between services and between the applications and services;
  - SOAP is typically on HTTP (others are possible)
  - UDDI and ebXML are used for service registration/publishing
- Web services use XML for representing input and data, as well as almost everything: SOAP, WSDL, UDDI, and ebXML are all in XML.
- Web services that use SOAP and WSDL are called "SOAP Services".

Definition of RESTful Service

- REST: Representational State Transfer
- A RESTful service is an **interface** that describes a **collection of resources** that are Web-accessible through standardized resource representations
  - URI (Uniformed Resource Identifier) is used to access the resource under the Web;
  - HTTP (Hypertext Transport Protocol) is used to exchange messages between services and between the applications and services;
  - RESTful services use one layer of protocol less (without SOAP), and HTTP is even more widely used.
- RESTful services use multiple resource representation languages for representing input and output data:
  - XML, HTML, XHTML, RSS Feed, Atom Feed, JSON
  - Does not deal with object types, and less related to OO

Service Protocol Stack